

ODETTE

Stereo repeater, from plucked strings to forty-second loops

BASIC HUMAN TASTES

Manual · v2.0.0 · October 2026



Odette repeats sound. SIZE picks a range of repeat times, from a couple of milliseconds, where repeats turn into strings and flangers, to forty seconds, where they become loops. TIME moves smoothly inside that range and bends the pitch like tape.

Everything is written into one long memory, so jumping from a small size to a large one brings back what was played earlier. TONE and BLUR shape the repeats a little more on every pass.

AT A GLANCE

- 8 nested sizes, from 1.3 ms to 42 s
- Smooth, pitch-bending TIME; clean jumps between sizes
- Clock sync with musical ratios, reverse, non-destructive freeze
- Stereo SPREAD and BOUNCE, tape-like DRIFT, and DUCK for live input
- 18 HP

CONTROLS

Screen	The eight sizes as a strip, the current one lit, and the current repeat time (or the clock ratio when CLOCK is patched).
SIZE	Range of repeat times; see the size table. Changing size jumps straight to the new time without bending the pitch.
TIME	Time between repeats within the size; to the right, longer. Turning it while repeats are sounding bends their pitch, like tape. With CLOCK patched it picks a ratio instead ($\times 2$ fully right, $1/1$ at the centre, $/2$ fully left) and changes jump cleanly.
FEEDBACK	Number of repeats, from one to endless and slightly beyond; saturation keeps it in check. While frozen, it chooses where in the memory the frozen loop starts.
TONE	Character of the repeats, applied again on every pass. Left: dark and warm, like an oil-can or bucket-brigade echo. Around three o'clock: neutral (default). Right: thin and crisp.
BLUR	Smears the repeats across the stereo field. Past half way, the smear is fed back into the loop and builds up over time.
SPREAD	Left and right repeat at different times, up to an octave apart in each direction.
MIX	Balance between the dry sound and the repeats. With MIX CV patched, the knob scales the CV.
DRIFT	A slow random wobble of the repeat time, different on each side, like tape.
DUCK	Lowers the repeats while sound is coming in and brings them back when it stops (fast attack, 250 ms release). Keeps a voice or an instrument clear.
SIZE · TIME · TONE	The three small knobs: amount and direction of the matching CV inputs.
REVERSE	Lit: repeats play backwards. In the smallest sizes, reversing tiny slices becomes a buzzy distortion.
FREEZE	Lit: nothing new comes in and the memory is left untouched. SIZE, TIME, FEEDBACK (as start point), TONE and BLUR keep acting on what you hear, without changing what is stored.
BOUNCE	Lit: the feedback crosses sides, so the repeats alternate left and right (ping-pong).

SIZES

Size	Repeat time	Typical use
1	1.3 to 20 ms	Plucked strings (with BEND), flanger
2	20 to 82 ms	Chorus, slapback, doubling
3	82 to 327 ms	Short echo
4	163 to 653 ms	Echo (default)
5	0.33 to 1.3 s	Long echo
6	0.65 to 2.6 s	Phrases
7	1.3 to 5.2 s	Phrases
8	2.6 to 42 s	Loops

INPUTS

SIZE · TIME · TONE	Scaled by their small knobs; 5 V covers the whole range.
FEEDBACK · BLUR	Added to their knobs; 5 V covers the whole range.
MIX	0 to 8 V, scaled by the MIX knob.
REVERSE · FREEZE	Each gate turns the matching button on or off.
CLOCK	Locks the repeats to a clock; TIME picks the ratio (see Clock sync).
BEND	Bends the repeat time. In size 1 it follows 1 V/oct, to play the repeats as a string. In other sizes it adds about 2 ms per volt, for chorus and vibrato, inverted on the right when SPREAD is off centre.
IN L · IN R	Audio inputs. IN R copies IN L when it has no cable.

OUTPUTS

OUT L · OUT R	Stereo output, ± 5 V.
PULSE	A trigger on every repeat, left or right. The light next to TIME flashes with it.

HOW IT WORKS

One memory for all sizes

Odette writes everything into one 64-second memory per side. The sizes are windows of different lengths onto that memory, so a short repeat played in size 2 is still there when you move to size 8, and repeats from smaller sizes leave traces in the larger ones.

Clock sync

With a clock in CLOCK (0.25 to 50 Hz), Odette measures its period and places the 1/1 ratio in the size that suits it best. TIME then steps through thirteen ratios from /2 to $\times 2$, and other sizes multiply or divide by powers of two. Changes jump in time without any pitch bend.

PATCH IDEAS

1. Tape echo

SIZE 4, FEEDBACK 60 %, TONE around eleven o'clock, DRIFT 30 %, BLUR 20 %.

2. Plucked string

SIZE 1, TIME fully left, FEEDBACK 85 %, MIX at the centre. Send a short noise burst or an Ernest hit into IN, and a pitch sequence into BEND.

3. Stereo chorus

SIZE 2, SPREAD a little off centre, FEEDBACK 30 %, a slow triangle LFO, turned down, into BEND.

4. Freeze and explore

Play into small sizes for a while, press FREEZE, move to size 8, then sweep FEEDBACK to travel through what was recorded.

5. A voice on stage

Microphone into IN, SIZE 5, FEEDBACK 50 %, DUCK 70 %: the echoes stay out of the way while the voice speaks and bloom in the pauses.

6. Back and forth

PULSE into REVERSE, sizes 3 to 5: the repeats turn round on every repeat.

SPECIFICATIONS

Width	18 HP
Repeat times	1.3 ms to 42 s in 8 sizes
Memory	64 s per side
Clock	0.25 to 50 Hz
Levels	Audio ± 5 V. PULSE 10 V, 1 ms. REVERSE, FREEZE and CLOCK react above about 2.5 V.